The Laplace Equation

Solution by Separation of Variables

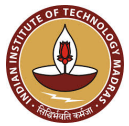
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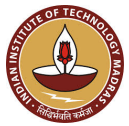
The Laplace Equation

- It arises in steady-state problems for the heat or wave equations that do not vary with time, so that $u_t = 0$ or $u_{tt} = 0$.
- In 2D:
$$\Delta u \equiv \nabla^2 u \equiv \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$
- In 3D:
$$\Delta u \equiv \nabla^2 u \equiv \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0.$$
- It plays an important role in gravitation, fluid dynamics, electrostatics, static deflection of a membrane, etc.
- It is the standard elliptic equation: $a = c = 1$, $b = 0$, $b^2 - 4ac = -4 < 0$.
- Example: the steady temperature in a thin plate satisfies $u_{xx} + u_{yy} = 0$ (put $u_t = 0$ in $u_t = k^2(u_{xx} + u_{yy})$).



Harmonic Functions

- A function $u \in C^2(D)$ with $\Delta u = 0$ in D is called harmonic in D.
- Examples: 1, x, y, xy, $x^2 - y^2$, $e^x \cos y$, $e^x \sin y$.
- In polar form: $r^n \cos n\theta$, $r^n \sin n\theta$, and (for $r > 0$) $\ln r$, $r^{-n} \cos n\theta$, $r^{-n} \sin n\theta$.
- The real and imaginary parts of an analytic function $f(z) = u + iv$ are harmonic.
- E.g. $z^2 = (x^2 - y^2) + i\,2xy$ and $e^z = e^x \cos y + i\, e^x \sin y$.
- Linear combinations of harmonic functions are harmonic (superposition).
- Harmonic functions are as smooth as possible: they are infinitely differentiable inside D.



Boundary Value Problems

- We consider $\Delta u = 0$ in a region D with conditions on its boundary ∂D; there is no initial condition.
- Dirichlet problem: u is prescribed on ∂D (e.g. temperature of the edges).
- Neumann problem: the normal derivative $\partial u/\partial n$ is prescribed on ∂D (heat flux).
- Mixed problem: u on part of ∂D, $\partial u/\partial n$ on the rest.
- We take D to be a rectangle, a strip, or a circular, semicircular, annular or wedge-shaped region.
- For such regions the boundary consists of coordinate lines (x, y = const or r, θ = const).
- This is exactly what is needed for the method of separation of variables.



Maximum Principle and Uniqueness

- Maximum principle. If u is harmonic in a bounded region D and continuous on $\overline{D}$, then

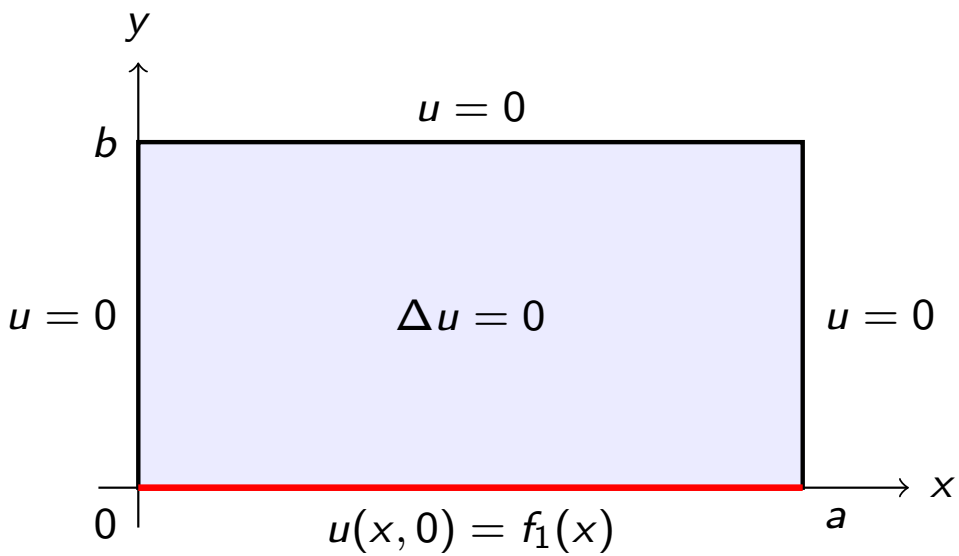
$$\min_{\partial D} u \leq u(x, y) \leq \max_{\partial D} u \quad \text{for all } (x, y) \in D.$$

- A steady temperature cannot be hotter inside the plate than on its edges.
- Uniqueness of the Dirichlet problem. If u_1, u_2 solve the same Dirichlet problem, then $w = u_1 - u_2$ is harmonic with $w = 0$ on ∂D.
- By the maximum principle, $0 \leq w \leq 0$ in D, so $u_1 = u_2$.
- Stability. If $|f_1 - f_2| \leq \varepsilon$ on ∂D, then $|u_1 - u_2| \leq \varepsilon$ in D.
- The Neumann problem is unique only up to an additive constant.



Rectangle, Case 1: Data on the Bottom Edge

- Let D be the rectangle $0 < x < a$, $0 < y < b$.



- Consider $u_{xx} + u_{yy} = 0$ with

$$u(0, y) = u(a, y) = u(x, b) = 0, \qquad u(x, 0) = f_1(x).$$

Case 1: Separation of Variables

- Let $u(x,y) = X(x)Y(y)$. Then $u_{xx} = X''Y$, $u_{yy} = XY''$, and $X''Y + XY'' = 0$:
$$\frac{X''}{X} = -\frac{Y''}{Y} = \mu \quad \text{(say)}.$$
- Hence
$$X'' - \mu X = 0, \tag{1}$$
$$Y'' + \mu Y = 0. \tag{2}$$
- $u(0,y) = X(0)Y(y) = 0$ and $u(a,y) = X(a)Y(y) = 0$ give, for a non-trivial solution,
$$X(0) = X(a) = 0. \tag{3}$$
- The two homogeneous conditions are in x, so we solve the eigenvalue problem for X first.



Case 1: The Eigenvalue Problem for X

- Consider the Sturm–Liouville problem (1), (3).
- $\mu = 0$: $X'' = 0 \Rightarrow X = Ax + B$; $X(0) = 0 \Rightarrow B = 0$;
  $X(a) = Aa = 0 \Rightarrow A = 0$. Trivial.
- $\mu = p^2 > 0$: $X = Ae^{px} + Be^{-px}$; $X(0) = 0 \Rightarrow B = -A$;
  $X(a) = A(e^{pa} - e^{-pa}) = 0 \Rightarrow A = 0$. Trivial.
- $\mu = -p^2 < 0$: $X = A\cos px + B\sin px$; $X(0) = 0 \Rightarrow A = 0$.
- $X(a) = 0 \Rightarrow B\sin pa = 0 = \sin n\pi$ for a non-trivial solution, so

$$\boxed{p = \frac{n\pi}{a}, \qquad X_n(x) = \sin\frac{n\pi x}{a}, \quad n \in \mathbb{N}}. \tag{4}$$

- (taking $B = 1$). Also $u(x, b) = X(x)Y(b) = 0 \Rightarrow Y(b) = 0$. (5)



Case 1: The Factor $Y(y)$

- For $\mu = -p^2 = -\frac{n^2\pi^2}{a^2}$, equation (2) becomes $Y'' - p^2 Y = 0$, so

$$Y(y) = C \cosh \frac{n\pi y}{a} + D \sinh \frac{n\pi y}{a}.$$

- It is easier to write the same general solution as

$$Y(y) = C \cosh \frac{n\pi(y-b)}{a} + D \sinh \frac{n\pi(y-b)}{a},$$

  because then $Y(b) = 0$ gives $C = 0$ directly.

- (With the first form: $C = -D \tanh \frac{n\pi b}{a}$, and $Y = \dfrac{D}{\cosh(n\pi b/a)} \sinh \dfrac{n\pi(y-b)}{a}$, the same result.).

- Hence

$$Y_n(y) = D_n \sinh \frac{n\pi(y-b)}{a}. \qquad (6)$$



Case 1: Solution

- By (4) and (6): $u_n(x,y) = X_n(x)Y_n(y) = D_n \sin \frac{n\pi x}{a} \sinh \frac{n\pi(y-b)}{a}$.
- By the principle of superposition,

$$u(x,y) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{a} \sinh \frac{n\pi(y-b)}{a}. \tag{7}$$

- Now $u(x,0) = f_1(x)$: $\sum_{n=1}^{\infty} \left[-D_n \sinh \frac{n\pi b}{a} \right] \sin \frac{n\pi x}{a} = f_1(x)$, so

$$-D_n \sinh \frac{n\pi b}{a} = \frac{2}{a} \int_0^a f_1(x) \sin \frac{n\pi x}{a} \, dx,$$

$$D_n = \frac{-2}{a \sinh(n\pi b/a)} \int_0^a f_1(x) \sin \frac{n\pi x}{a} \, dx. \tag{8}$$

- Equivalently, with $c_n = \frac{2}{a} \int_0^a f_1 \sin \frac{n\pi x}{a} \, dx$:

$$u(x,y) = \sum_{n=1}^{\infty} c_n \frac{\sinh\left(n\pi(b-y)/a\right)}{\sinh(n\pi b/a)} \sin \frac{n\pi x}{a}.$$

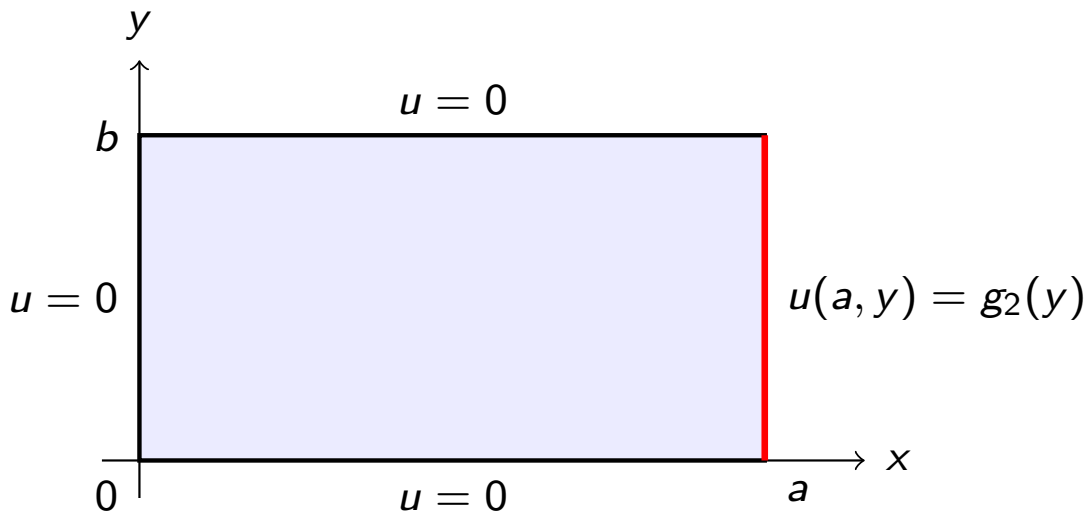


Remarks on the Rectangle Solution

- The factor $\dfrac{\sinh(n\pi(b-y)/a)}{\sinh(n\pi b/a)}$ equals 1 on $y=0$ and 0 on $y=b$.
- For large n it behaves like $e^{-n\pi y/a}$: high-frequency boundary data decay quickly away from the edge.
- So u is smooth inside even if f_1 has corners or jumps.
- The x-direction carries the sines (oscillation), the y-direction carries the hyperbolic functions (growth/decay).
- Separating with the opposite sign of μ would put sines in y: the choice is dictated by where the homogeneous conditions are.
- Rule: the variable with two homogeneous boundary conditions gets the eigenvalue problem.



Rectangle, Case 2: Data on the Right Edge



- Here $u_{xx} + u_{yy} = 0$, $u(x, 0) = u(x, b) = 0 = u(0, y)$, $u(a, y) = g_2(y)$.
- With $u = XY$: $X'' - \mu X = 0$ (1), $Y'' + \mu Y = 0$ (2).
- $u(x, 0) = X(x)Y(0) = 0$ and $u(x, b) = X(x)Y(b) = 0$ give $Y(0) = 0$, $Y(b) = 0$. (3)
- Now the homogeneous conditions are in y, so we solve the eigenvalue problem (2)–(3).



Case 2: Eigenfunctions and Solution

- For $\mu = 0$ and $\mu = -p^2$: $Y \equiv 0$ (exercise).
- For $\mu = p^2$: $Y = A\cos py + B\sin py$; $Y(0) = 0 \Rightarrow A = 0$; $Y(b) = 0 \Rightarrow p = \dfrac{n\pi}{b}$.

$$Y_n(y) = \sin\frac{n\pi y}{b}, \quad n \in \mathbb{N}. \tag{4}$$

- $u(0, y) = X(0)Y(y) = 0 \Rightarrow X(0) = 0.$ (5)
- From (1): $X'' - \frac{n^2\pi^2}{b^2}X = 0 \Rightarrow X = C\cosh\frac{n\pi x}{b} + D\sinh\frac{n\pi x}{b}$; by (5), $C = 0$:

$$X_n(x) = D_n\sinh\frac{n\pi x}{b}. \tag{6}$$

- By superposition,

$$u(x, y) = \sum_{n=1}^{\infty} D_n\sinh\frac{n\pi x}{b}\ \sin\frac{n\pi y}{b}. \tag{7}$$



Case 2: Coefficients

- Now $u(a, y) = g_2(y)$:

$$\sum_{n=1}^{\infty} D_n \sinh \frac{n\pi a}{b} \sin \frac{n\pi y}{b} = g_2(y).$$

- This is a Fourier sine series on $[0, b]$, so

$$D_n \sinh \frac{n\pi a}{b} = \frac{2}{b} \int_0^b g_2(y) \sin \frac{n\pi y}{b} \, dy,$$

$$\boxed{D_n = \frac{2}{b \sinh(n\pi a/b)} \int_0^b g_2(y) \sin \frac{n\pi y}{b} \, dy.} \quad (8)$$

- The desired solution is given by (7) and (8).
- The ratio $\sinh(n\pi x/b)/\sinh(n\pi a/b)$ equals 1 on the edge $x = a$ and decays towards $x = 0$.



Example 1: $g_2(y) = 100$

- One edge of the rectangle is kept at 100°, the other three at 0°.
- From (8):

$$D_n = \frac{2}{b\sinh(n\pi a/b)} \int_0^b 100 \sin \frac{n\pi y}{b}\, dy = \frac{200}{b\sinh(n\pi a/b)} \left[-\frac{b}{n\pi} \cos \frac{n\pi y}{b} \right.$$

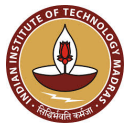
- So

$$D_n = \begin{cases} \dfrac{400}{n\pi \sinh(n\pi a/b)}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}.$$

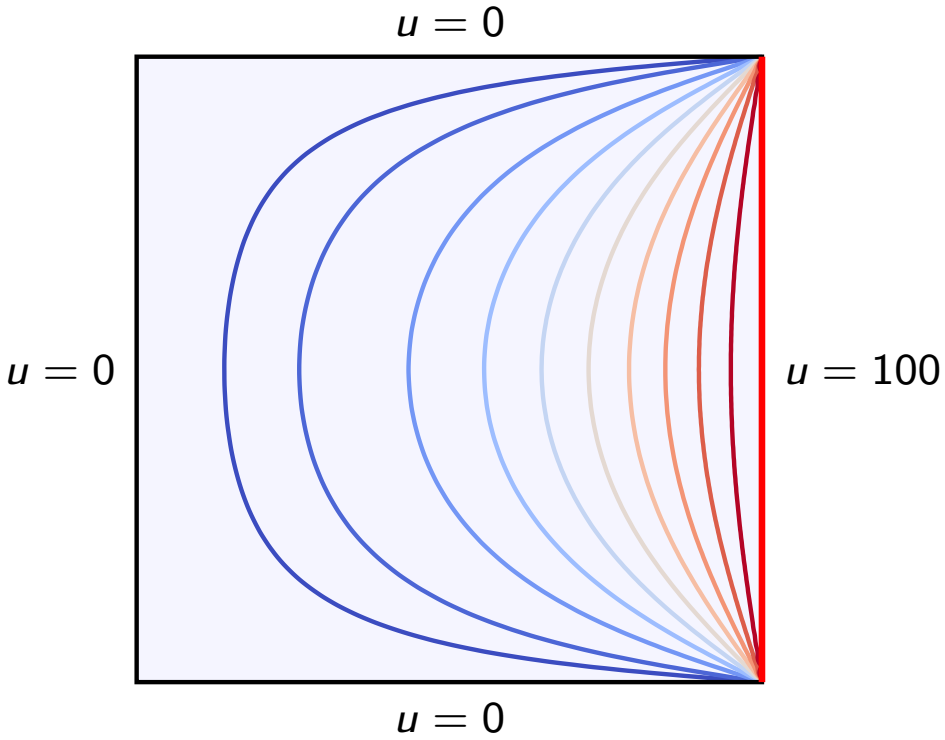
- Hence

$$u(x,y) = \frac{400}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \frac{\sinh\left((2n-1)\pi x/b\right)}{\sinh\left((2n-1)\pi a/b\right)} \sin \frac{(2n-1)\pi y}{b}.$$

- At the corners $(a, 0)$ and (a, b) the data jump from 100 to 0; the series gives the average 50.



Picture: Isotherms for Example 1 ($a = b = 1$)



- Level curves $u = 5, 10, 20, \ldots, 90$ (blue = cold, red = hot).
- All isotherms end at the two hot corners; heat flows across them from the hot edge.

Cases 3 and 4

- Case 3: $u(x,b) = f_2(x)$ on the top edge, $u = 0$ on the other three edges.
- As in Case 1, $X_n = \sin \frac{n\pi x}{a}$ and now $Y(0) = 0$ gives $Y_n = \sinh \frac{n\pi y}{a}$:

$$u = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{a} \sinh \frac{n\pi y}{a}, \quad D_n = \frac{2}{a \sinh \frac{n\pi b}{a}} \int_0^a f_2 \sin \frac{n\pi x}{a}\, dx.$$

- Case 4: $u(0,y) = g_1(y)$ on the left edge, $u = 0$ on the other three edges.
- As in Case 2, $Y_n = \sin \frac{n\pi y}{b}$ and $X(a) = 0$ gives $X_n = \sinh \frac{n\pi(a-x)}{b}$:

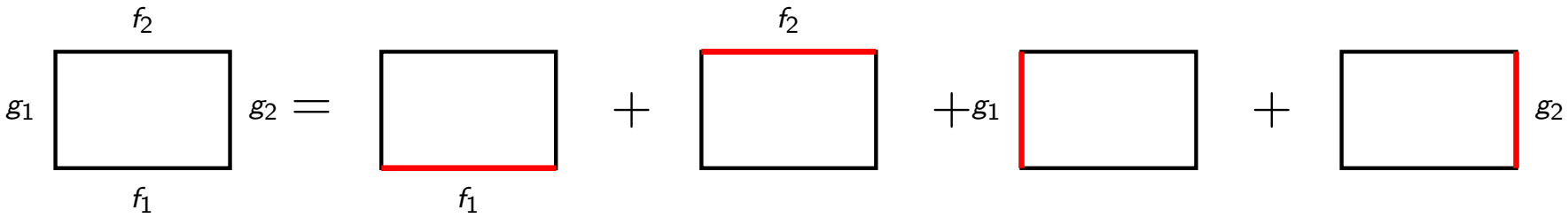
$$u = \sum_{n=1}^{\infty} D_n \sinh \frac{n\pi(a-x)}{b} \sin \frac{n\pi y}{b}, \quad D_n = \frac{2}{b \sinh \frac{n\pi a}{b}} \int_0^b g_1 \sin \frac{n\pi y}{b}\, dy.$$

- (Derive these as homework, following Cases 1 and 2.).



General Case: Superposition of Four Problems

- Consider $u_{xx} + u_{yy} = 0$ with $u(x, 0) = f_1(x)$, $u(x, b) = f_2(x)$, $u(0, y) = g_1(y)$, $u(a, y) = g_2(y)$.



- Split the problem into four cases, each with data on one edge only and zero on the others.
- Solve each by Cases 1–4 and use the principle of superposition: $u = \sum_{i=1}^{4} u_i$.
- If insulated (Neumann) or mixed boundary conditions are prescribed, use the corresponding eigenvalue problems for x or y (cosines, quarter-wave sines, etc.).

Example 2: Two Hot Edges

- Solve $\Delta u = 0$ in $0 < x < \pi$, $0 < y < \pi$ with $u(x, \pi) = \sin x$, $u(\pi, y) = \sin 2y$, and $u = 0$ on the other edges.
- Top edge (Case 3, $a = b = \pi$): only $n = 1$ appears, $D_1 \sinh \pi = 1$:

$$u_1 = \frac{\sinh y}{\sinh \pi} \sin x.$$

- Right edge (Case 2): only $n = 2$, $D_2 \sinh 2\pi = 1$:

$$u_2 = \frac{\sinh 2x}{\sinh 2\pi} \sin 2y.$$

- Hence

$$u(x, y) = \frac{\sinh y \, \sin x}{\sinh \pi} + \frac{\sinh 2x \, \sin 2y}{\sinh 2\pi}.$$

- Check: each term is harmonic, e.g. $(\sinh y \sin x)_{xx} + (\sinh y \sin x)_{yy} = -\sinh y \sin x + \sinh y \sin x = 0$.



Homework: Rectangle

- (1) $\Delta u = 0$ in $0 < x < a$, $0 < y < b$; $u_x(0, y) = u_x(a, y) = 0$, $u(x, 0) = 0$, $u(x, b) = \alpha x$.

  Hint: $X_n = \cos \frac{n\pi x}{a}$, $n \geq 0$; for $n = 0$ take $Y_0 = y$.

- (2) $\Delta u = 0$; $u_x(0, y) = 0$, $u(a, y) = 0$, $u(x, b) = 0$,

$$u(x, 0) = \begin{cases} 1, & 0 < x < a/2, \\ -1, & a/2 < x < a. \end{cases}$$

  Hint: $X_n = \cos \frac{(2n-1)\pi x}{2a}$.

- (3) $\Delta u = 0$; $u_y(x, 0) = u_y(x, b) = 0$, $u(a, y) = 0$,

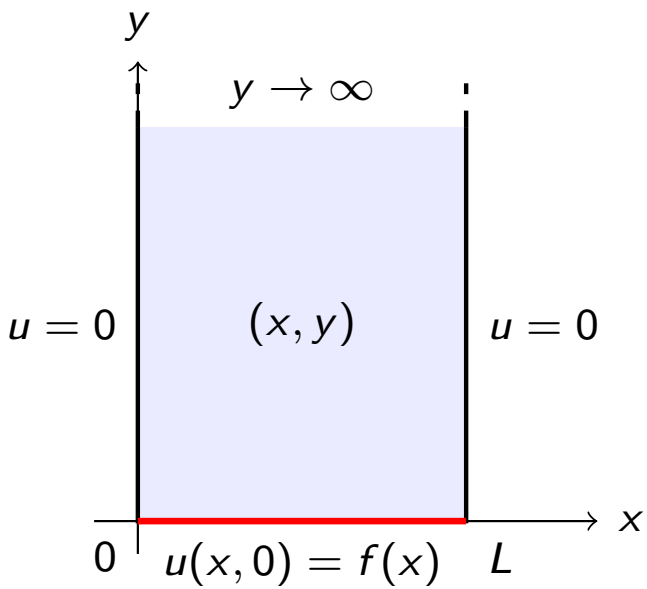
$$u(0, y) = \begin{cases} y, & 0 < y < b/2, \\ b - y, & b/2 < y < b. \end{cases}$$

- (4) Derive Cases 3 and 4.
- (5) Solve $\Delta u = 0$ in the unit square with $u = 1$ on all four edges, first directly and then by superposition; compare.



One-sided Infinitely Long Plate

- An infinitely long uniform plate is bounded by two parallel edges $x = 0$ and $x = L$.
- The breadth of the edge $y = 0$ is L, and this edge is maintained at a temperature $f(x)$.
- All the other edges are maintained at temperature zero. Find the steady-state temperature at any interior point.



- In the steady state, $u_t = k^2(u_{xx} + u_{yy})$ reduces to $u_{xx} + u_{yy} = 0$.

Strip: Boundary Conditions and Separation

- The boundary conditions are

$$u(0,y) = 0, \quad u(L,y) = 0, \qquad 0 < y < \infty,$$

$$u(x,\infty) = 0, \quad u(x,0) = f(x), \qquad 0 < x < L.$$

- Let $u = X(x)Y(y)$: $\dfrac{X''}{X} = -\dfrac{Y''}{Y} = \mu$, $X'' - \mu X = 0$ (1), $Y'' + \mu Y = 0$ (2).
- $u(0,y) = 0 = u(L,y) \Rightarrow X(0) = 0 = X(L).$ (3)
- As for the rectangle, only $\mu = -p^2$ gives non-trivial solutions:

$$p = \frac{n\pi}{L}, \qquad X_n(x) = \sin\frac{n\pi x}{L}, \quad n \in \mathbb{N}.$$

- From (2): $Y'' - \frac{n^2\pi^2}{L^2}Y = 0 \Rightarrow Y = C\,e^{n\pi y/L} + D\,e^{-n\pi y/L}$.



Strip: Solution

- $u(x, \infty) = 0 \Rightarrow Y(\infty) = 0$ for a non-trivial solution.
- Since $e^{n\pi y/L} \to \infty$ while $e^{-n\pi y/L} \to 0$, we need $C = 0$:

$$Y_n(y) = D_n\, e^{-n\pi y/L}.$$

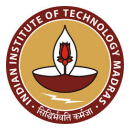
- By superposition,

$$u(x, y) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{L}\, e^{-n\pi y/L}. \tag{4}$$

- From $u(x, 0) = f(x)$:

$$D_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L}\, dx. \tag{5}$$

- Using (5) in (4), we get the desired solution.
- On a bounded rectangle we had sinh; on the infinite strip the decaying exponential replaces it.

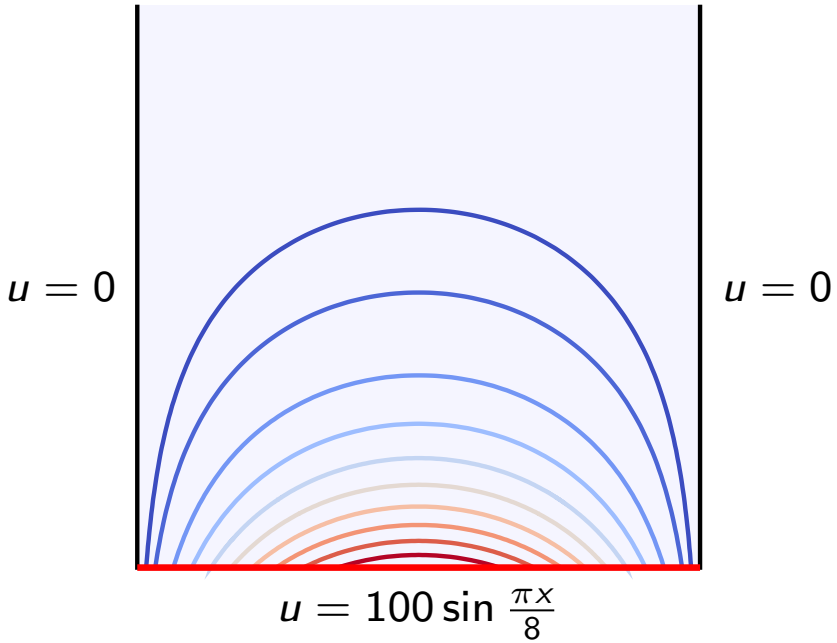


Example 3: $u(x,0) = 100 \sin \frac{\pi x}{8}$

- Solve $\Delta u = 0$ with $u(0,y) = u(8,y) = 0$ $(0 < y < \infty)$, $u(x,\infty) = 0$, $u(x,0) = 100 \sin \frac{\pi x}{8}$ $(0 < x < 8)$.
- Here $L = 8$: $u(x,y) = \sum_{n=1}^{\infty} D_n e^{-n\pi y/8} \sin \frac{n\pi x}{8}$.
- $u(x,0) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{8} = 100 \sin \frac{\pi x}{8}$.
- Comparing: $D_1 = 100$ and $D_n = 0$ for $n \neq 1$.
- Hence
$$u(x,y) = 100\, e^{-\pi y/8} \sin \frac{\pi x}{8}.$$
- Check: $u_{xx} = -\frac{\pi^2}{64} u$ and $u_{yy} = \frac{\pi^2}{64} u$, so $\Delta u = 0$.



Picture: Isotherms for Example 3



- Level curves $u = 5, 10, 20, \ldots, 90$: arches over the hot edge, shrinking as y grows.
- The temperature falls off exponentially with the distance from the hot edge.

Example 4: Triangular Edge Temperature

- A rectangular plate with insulated surface is 10 cm wide and so long compared to its width that it may be considered infinite.
- The temperature at the short edge $y = 0$ is $u(x, 0) = 20x$ for $0 \le x \le 5$ and $20(10 - x)$ for $5 \le x \le 10$; the long edges are kept at 0°C. Find the steady-state temperature.
- With $L = 10$: $u(x, y) = \sum D_n \sin \frac{n\pi x}{10} e^{-n\pi y/10}$, and

$$D_n = \frac{2}{10}\left[\int_0^5 20x \sin\frac{n\pi x}{10}\,dx + \int_5^{10} 20(10-x)\sin\frac{n\pi x}{10}\,dx\right] = \frac{800}{n^2\pi^2}\sin\frac{n\pi}{2}$$

- Hence

$$u(x, y) = \frac{800}{\pi^2}\sum_{n=1}^{\infty}\frac{\sin(n\pi/2)}{n^2}\sin\frac{n\pi x}{10}\,e^{-n\pi y/10}.$$



Example 4 with Insulated Long Edges

- Suppose now the long edges are insulated: $u_x(0,y) = 0 = u_x(10,y)$, with the same $u(x,0)$.
- Then $X_n = \cos\frac{n\pi x}{10}$, $n = 0,1,2,\ldots,$ and for $n = 0$: $Y_0'' = 0 \Rightarrow Y_0 = c + dy$.
- Boundedness as $y \to \infty$ forces $d = 0$, so the constant mode does not decay:

$$u(x,y) = A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{10}\, e^{-n\pi y/10}, \qquad A_0 = \frac{1}{10}\int_0^{10} u(x,0)\,dx = 50.$$

- Hence $u \to 50$ as $y \to \infty$; the condition $u(x,\infty) = 0$ must be replaced by boundedness.
- Physically: heat entering at $y = 0$ cannot escape through insulated sides.
- $A_n = \frac{2}{10}\int_0^{10} u(x,0)\cos\frac{n\pi x}{10}\,dx$ (homework: show $A_n = 0$ for odd n)

